

TECHNICAL SUBMISSION

Advanced IoT-Based Particulate Matter and Air Quality Monitoring System Using ESP32 with Real-Time Analytics

Final reproducibility, implementation, data, verification and evidence package

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1. Submission Scope

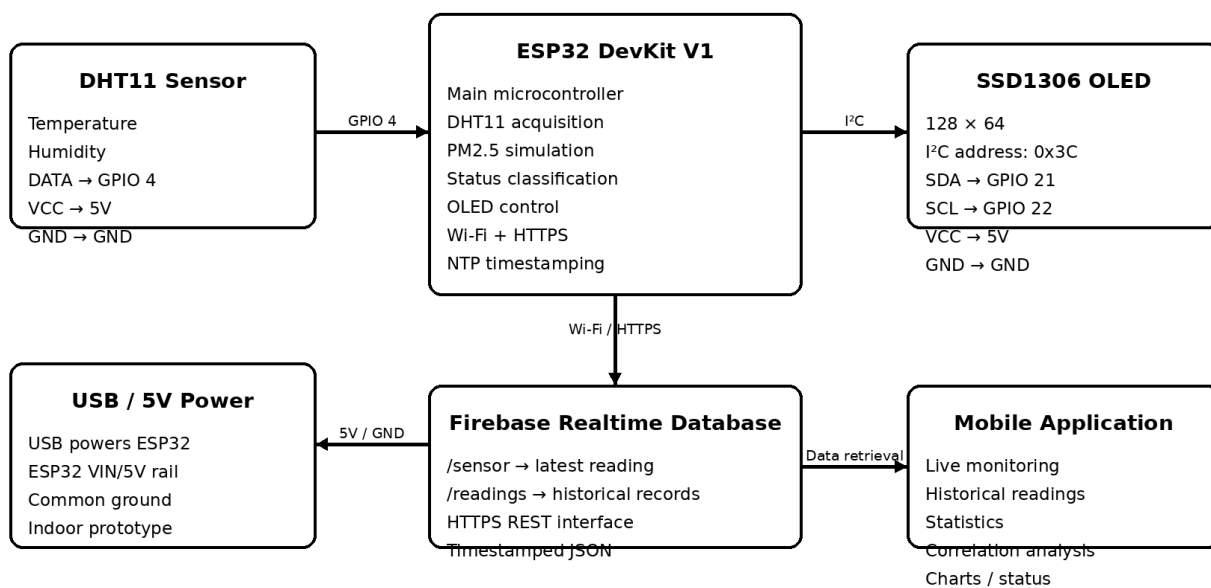
This document accompanies the reproducible technical submission package and is built from the implemented ESP32 firmware, Flutter application source, Firebase export and supplied evidence. Numerical results are recalculated from the exact Firebase export.

2. Implemented System

DHT11 acquisition → ESP32 processing → SSD1306 OLED → Wi-Fi/HTTPS → Firebase Realtime Database → Flutter mobile dashboard and historical analytics.

ESP32 IoT Air-Quality Monitoring System - Implemented Prototype

Actual hardware path: ESP32 + DHT11 + SSD1306 OLED; PM2.5 is software-simulated



PM2.5 channel: firmware simulation only (no physical particulate-matter sensor integrated)

Measurement cycle: 5 s | OLED: 128×64 I2C | DHT11: GPIO 4 | SDA: GPIO 21 | SCL: GPIO 22

Figure 1. Final implemented system schematic. PM2.5 is software-simulated in the current firmware.

3. Hardware and Pin Configuration

Component	Interface / Pin	Implementation
ESP32 DevKit V1	Main controller	Acquisition, processing, Wi-Fi/HTTPS
DHT11	DATA → GPIO 4	Real temperature/humidity
SSD1306	SDA 21; SCL 22; 0x3C	Local I2C display
Power	USB / 5V input	Prototype power

Measured supply current was not provided, so no current-consumption value is claimed.

4. Raw Dataset and Data Quality

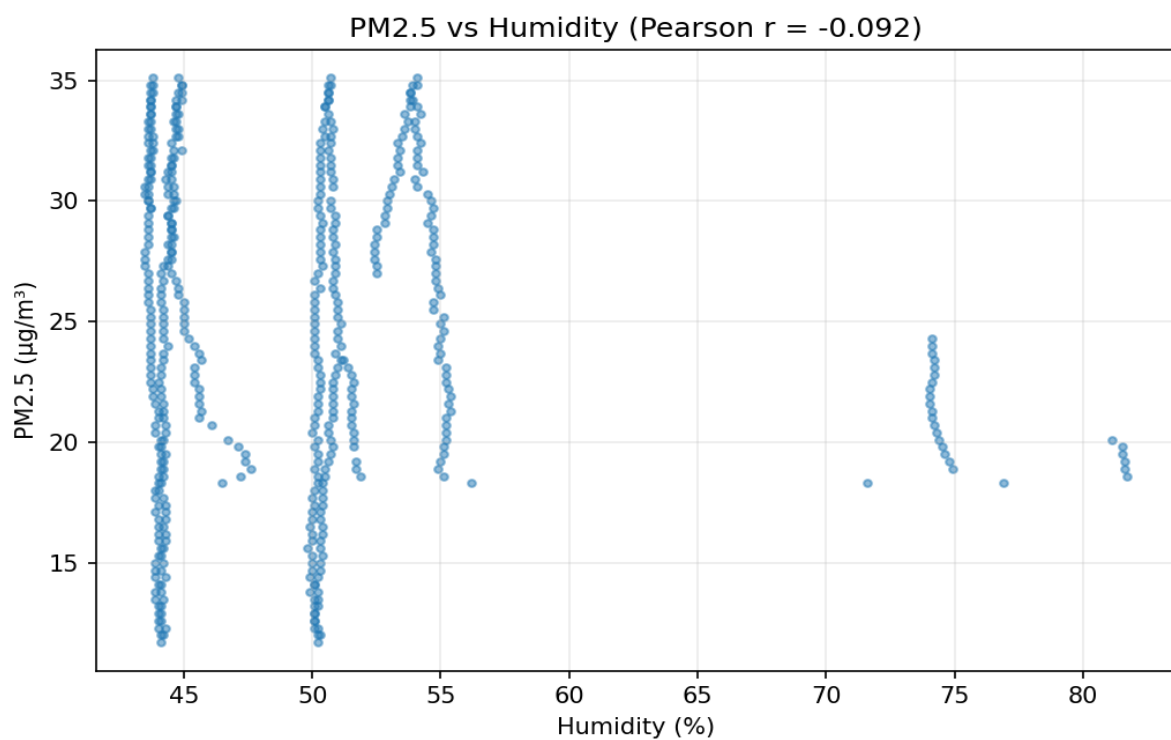
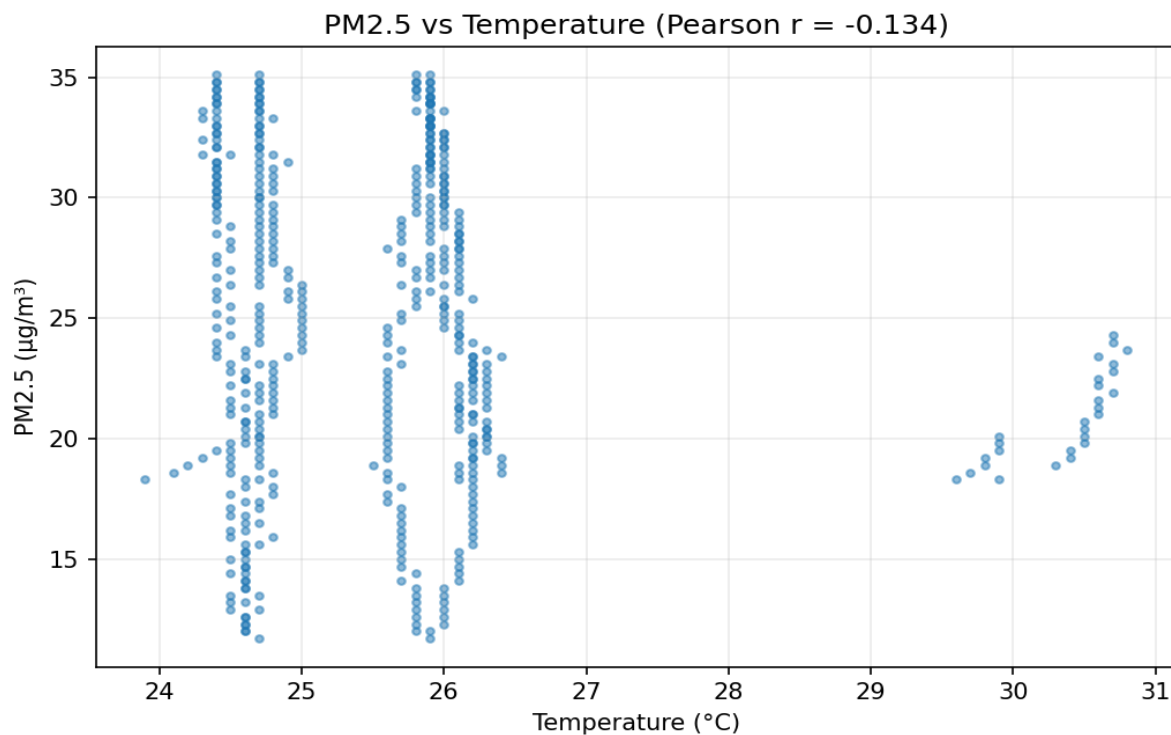
The Firebase export contains 510 historical records: 424 valid explicit timestamps, 84 missing timestamp fields and 2 TIME_NOT_SYNCED records.

5. Processed Numerical Results

Metric	Min	Mean	Max	Pop. SD	Unit
PM2.5	11.7	24.595	35.1	6.385	µg/m³
Temperature	23.9	25.588	30.8	1.320	°C
Humidity	43.5	49.678	81.7	7.340	%

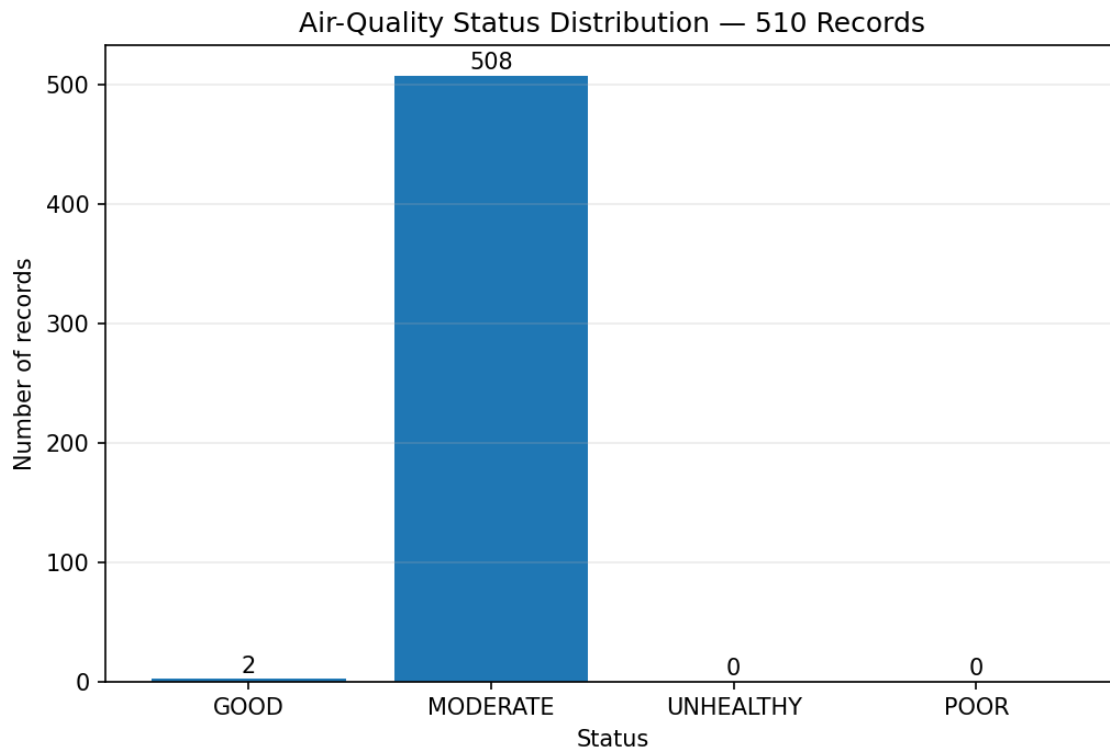
6. Correlation Analysis

PM2.5 vs Temperature: $r = -0.134$. PM2.5 vs Humidity: $r = -0.092$.



7. Status Distribution

Status	Count	%
GOOD	2	0.39
MODERATE	508	99.61
UNHEALTHY	0	0.00
POOR	0	0.00



8. Verification and Validation

T01 — DHT11 acquisition: PASS – functional evidence; formal accuracy not validated

T02 — OLED display: PASS – functional

T03 — PM2.5 simulation: DEVIATION – observed 11.7–35.1 $\mu\text{g}/\text{m}^3$; ± 0.3 step crosses bounds by one step

T04 — Latest Firebase upload: PASS

T05 — Historical Firebase logging: PASS – export evidence

T06 — Timestamp generation: PASS with limitation – 424 valid timestamps; 84 missing and 2 unsynced

T07 — Mobile data retrieval: PASS – functional evidence

T08 — Statistical analysis: PASS – 510 records

T09 — Correlation analysis: PASS – $r=-0.134$ (temperature), $r=-0.092$ (humidity)

T10 — Wi-Fi reconnection: PENDING

T11 — Long-duration uptime: PENDING

T12 — Sensor accuracy: PENDING

T13 — Security hardening: PENDING – development configuration uses placeholder credentials and `setInsecure()`

9. Engineering Deviations

The current PM2.5 simulation produces observed boundary values 11.7 and 35.1 $\mu\text{g}/\text{m}^3$ because the ± 0.3 update occurs before the direction reversal. The deviation is recorded and a corrective clamp is recommended. Production TLS certificate validation and Firebase rule hardening are also pending.

10. Reproducibility and Security

The package contains credential-sanitized ESP32 source, Flutter source, raw/processed data, schematic, charts and verification tables. The original Wi-Fi password is not included. The firmware's development TLS bypass is explicitly documented.